

IDEATION DOCUMENT

AI-Powered Internal Knowledge Discovery System (LOCAL AI)

1. Problem Statement

Modern organisations generate a vast number of knowledge artifacts including design documents, reports, technical notes, and project descriptions. Despite the abundance of information, knowledge accessibility remains inefficient.

Organisational knowledge is often:

- Distributed across storage systems
- Buried within large documents
- Difficult to search semantically
- Dependent on manual discovery
- Prone to duplication

This leads to recurring operational challenges:

- ✗ Repeated work
 - ✗ Knowledge invisibility
 - ✗ Onboarding friction
 - ✗ Increased cognitive load
 - ✗ Inefficient knowledge utilisation
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2. Motivation / Why This System Exists

Traditional document systems focus on **file retrieval**, requiring users to know:

- What document exists
- Where it is stored
- What terminology was used

However, users typically operate with **knowledge intent**, not document intent.

Example mismatch:

User Need → “Has this already been designed?”

System Expectation → “Search exact file name”

This misalignment causes knowledge discovery friction.

The proposed system exists to transform passive document repositories into an **active semantic knowledge discovery engine**.

3. Core System Philosophy

The system redefines knowledge interaction by supporting **two distinct operational modes** based on user intent.

Instead of treating all queries uniformly, the system differentiates between:

👉 **Knowledge Discovery Intent**

👉 **Knowledge Understanding Intent**

This eliminates unnecessary reasoning overhead and improves response correctness.

4. Target Users

Primary users include:

 Developers / Engineers

 Team Members

 Interns / New Joiners

 Stakeholders

Different users exhibit different knowledge needs.

5. Key Challenges Identified

1. Knowledge Discovery Friction

Users struggle to confirm existence of prior knowledge.

2. Semantic Search Limitations

Keyword systems fail for conceptual queries.

3. Knowledge Drift

Static documents become outdated.

4. Knowledge Duplication

Teams unknowingly recreate solutions.

5. Onboarding Inefficiency

New members lack visibility into ongoing initiatives.

6. Proposed Solution Vision

The system introduces an **AI-assisted semantic knowledge discovery framework** that:

- ✓ Enables semantic retrieval
 - ✓ Differentiates user intent
 - ✓ Minimises hallucination risk
 - ✓ Supports deterministic responses
 - ✓ Provides dynamic knowledge synthesis
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7. System Behaviour Modes

✓ **MODE 1 — Knowledge Discovery / Existence Check**

User Intent

User seeks confirmation of knowledge presence.

Example Queries:

- “Does authentication design exist?”
 - “Is there document on caching strategy?”
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System Behaviour

Pipeline:

Query → Intent Validation → Retrieval Engine → Deterministic Output

Output Style

- ✓ Yes / No
- ✓ Relevant Document IDs

Example Response:

“Yes, relevant knowledge exists.”

Value Delivered

- ✓ Eliminates unnecessary reasoning
 - ✓ Reduces hallucination risk
 - ✓ Improves response precision
 - ✓ Enables fast knowledge navigation
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✓ **MODE 2 — Knowledge Understanding / Project Overview**

User Intent

User seeks conceptual understanding or synthesis.

Example Queries:

- “Explain Trinetra project.”
 - “What initiatives are ongoing?”
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System Behaviour

Pipeline:

Query → Intent Validation → Retrieval Engine → Knowledge Aggregation → Model Synthesis

Output Style

- ✓ Structured Explanation
- ✓ High-Level Overview
- ✓ Grounded Response

Example Response:

“Trinetra is an internal analytics platform designed to...”

Value Delivered

- ✓ Reduces onboarding friction
- ✓ Improves knowledge visibility
- ✓ Supports contextual understanding
- ✓ Enables knowledge synthesis

8. High-Level System Architecture

The system follows a **three-module architecture**:

Module 1 — Document Ingestion Pipeline

Transforms raw documents → semantic memory units.

Purpose:

 Knowledge Integrity Engine

Module 2 — Retrieval Engine

Extracts semantically relevant knowledge based on intent.

Purpose:

 Relevance & Signal Extraction Engine

Module 3 — Knowledge Delivery Layer

Transforms retrieved knowledge → grounded responses.

Purpose:

 Interpretation & Trust Engine

9. Expected Organisational Impact

Operational Benefits

- ✓ Reduced duplicated work
 - ✓ Improved knowledge utilisation
 - ✓ Faster discovery cycles
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✓ Human-Centric Benefits

- ✓ Reduced cognitive load
 - ✓ Improved onboarding
 - ✓ Enhanced productivity
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🎯 10. Strategic Advantages

- ✓ Semantic retrieval vs keyword retrieval
 - ✓ Intent-aware behaviour vs uniform responses
 - ✓ Deterministic discovery vs hallucinated answers
 - ✓ Dynamic synthesis vs static lookup
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🎯 11. Critical Design Principles

- ✓ Knowledge Integrity First
 - ✓ Retrieval Quality Over Model Choice
 - ✓ Intent-Aware Behaviour
 - ✓ Noise Suppression Discipline
 - ✓ Grounded Response Enforcement
 - ✓ Explainability & Trust Preservation
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🎯 12. Risks & Constraints

Recognised risks:

- ✗ Knowledge drift
- ✗ Retrieval noise
- ✗ Semantic similarity illusion
- ✗ Hallucination risk

Mitigation Strategy:

- 👉 Dynamic retrieval + grounding constraints.
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